

Experiment - 19

High Level Programming Extensions – Triggers

```
MySQL 8.0 Command Line Client
mysql> USE ProcedureDB;
Database changed
mysql>
mysql> CREATE TABLE Student_Audit (
  ->   AuditID INT AUTO_INCREMENT PRIMARY KEY,
  ->   StudentID INT,
  ->   StudentName VARCHAR(50),
  ->   ActionType VARCHAR(20),
  ->   ActionTime TIMESTAMP DEFAULT CURRENT_TIMESTAMP
  -> );
Query OK, 0 rows affected (0.13 sec)

mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER AfterStudentInsert
  -> AFTER INSERT ON Student
  -> FOR EACH ROW
  -> BEGIN
  ->   INSERT INTO Student_Audit
  ->   (StudentID, StudentName, ActionType)
  ->   VALUES
  ->   (NEW.StudentID, NEW.StudentName, 'INSERT');
  -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> INSERT INTO Student
  -> VALUES (105, 'Priya', 'IT', 88);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Student;
+-----+-----+-----+-----+
| StudentID | StudentName | Department | Marks |
+-----+-----+-----+-----+
| 101 | Indhu | CSE | 85 |
| 102 | Meghana | IT | 78 |
| 103 | Shalini | ECE | 92 |
| 104 | Teju | CSE | 67 |
| 105 | Priya | IT | 88 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT * FROM Student_Audit;
+-----+-----+-----+-----+-----+
| AuditID | StudentID | StudentName | ActionType | ActionTime |
+-----+-----+-----+-----+-----+
| 1 | 105 | Priya | INSERT | 2026-08-11 17:35:44 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> ALTER TABLE Student_Audit
-> ADD OldMarks INT,
-> ADD NewMarks INT;
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER AfterStudentUpdate
-> AFTER UPDATE ON Student
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO Student_Audit
->     (StudentID, StudentName, ActionType, OldMarks, NewMarks)
->     VALUES
->     (NEW.StudentID, NEW.StudentName, 'UPDATE',
->     OLD.Marks, NEW.Marks);
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> UPDATE Student
-> SET Marks = 95
-> WHERE StudentID = 105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
mysql> SELECT * FROM Student_Audit;
+-----+-----+-----+-----+-----+-----+-----+
| AuditID | StudentID | StudentName | ActionType | ActionTime | OldMarks | NewMarks |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | 105 | Priya | INSERT | 2026-08-11 17:35:44 | NULL | NULL |
| 2 | 105 | Priya | UPDATE | 2026-08-11 17:36:09 | 88 | 95 |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```